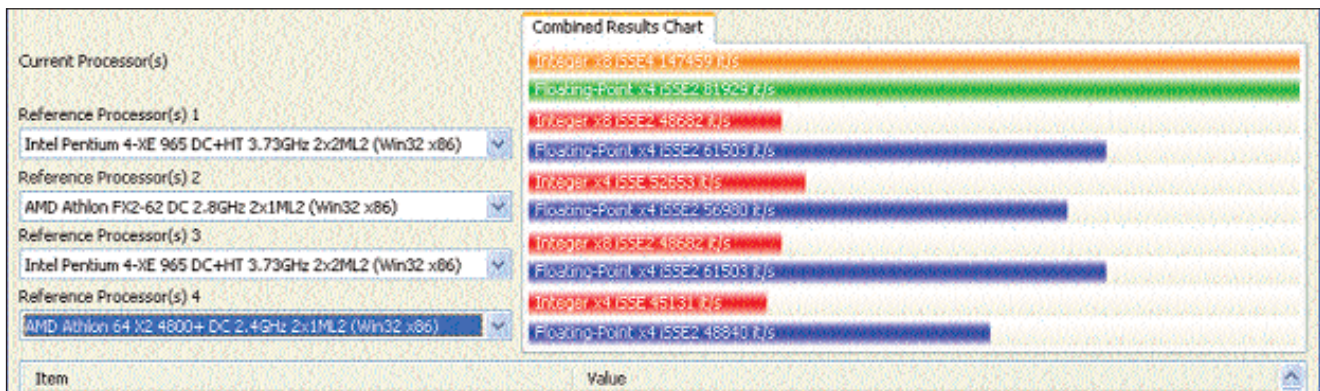




SiSoft Sandra 2007 CPU multimedia benchmark. The difference is alarming!

Core Duo mobile range of processors, along with, of course, a number of innovations. What emerges is a completely new product, a combination of the best, a regression of the worst.

Let's take a look at the Core 2 family of processors, and also compare them with a couple of older processors.

Intel has continued with their 65 nm fab, but the biggest improvement we see is in the steep reduction of die size, meaning Intel can churn out Conroe's much cheaper than their older-generation processors!

The Benchmarks

Our Core 2 Extreme booted into a speed of 1.6 GHz. This came as a surprise, but we realised that the multiplier is unlocked (only the Extreme version has this feature). The processor can throttle down to 1.6 GHz (the multiplier being 6, the FSB being 266, quad-pumped). When needed, the multiplier will shoot up to 11, which is its maximum value.

To test the Conroe, we used the Intel 975XBX "Bad Axe" motherboard, based on the Glenwood

975 Express chipset. We used 2 GB of memory (dual-channel configured)—Corsair's XMS 2 DDR2 800 MHz timed at 4-4-4-15. The graphics card was an Asus X1900XTX. All tests were performed at these timings with no overclocking whatsoever. As you'll see, Conroe doesn't really need it!

Super Pi—a very well-known CPU benchmark—was run first. Its 1M test completed in 17.593 seconds, a new record in our labs. The AMD 64 5000+ took 33.110 seconds for the same test. Amazing: nearly *double* the performance for the Conroe.

We then ran Future Mark's PC Mark 05; here the Conroe scored 7649. The CPU benchmark read 7273; the AMD 64 5000+ at 2.6 GHz gives just 5348, another victory for Conroe. However the Core 2 X6800 is targeted at the AMD FX 62 and not the 5000+, which is a couple of notches lower on AMD's AM2 processor line-up. Interesting days to follow, lets play the wait and watch game!

Our third benchmark was the Multimedia Instruction test under SiSoft Sandra 2007. We've

decided to let the pretty picture above speak for itself.

Look impressive? We were as awestruck as you probably are right now! Victory, and how! Strike three!

Let's look at a game benchmark now. We ran *Far Cry* with the version 1.32 patch installed, which is one of the more CPU-intensive games around, and which scales very well with hardware changes. We used the same memory and graphics card on both platforms. The motherboard we used to test the 5000+ was the ASUS M5N SLI 32, based around NVIDIA's latest nForce 590 SLI chipset.

Worth The Hype...? Yes!

That's the answer! The Conroe was well worth the wait. The Pentium 4 and its follies seem a distant memory now, and at least this time round, Intel seems to have gotten things very right. The Conroe becomes the fastest thing that we have laid eyes on, and we're deadly serious. Unlike most platform upgrades which improve performance by narrow margins—AMD's AM2 VS 939 is a good example—the Conroe is seemingly millennia ahead!

These are little more than initial reactions at best, but some where down the line, things can only get rosier for Intel. We're anxiously awaiting the FX 62, so expect a very comprehensive benchmark of both these heavyweights very, very soon.

The real winners, though, are us, the users. With the lower versions of the Conroe promising very competitive rates (which will push down the rates of older-generation processors even more), the time for an upgrade for all those hopefuls seems now!

Hold on a bit, though—Conroe's will take a little time to become readily available. ■

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The Core 2 Family, Predecessors and the competition

Processor Model Number	Core (GHz)	Bus Speeds	L2 Cache	Mfg. process(nm)	Transistors (millions)
INTEL					
Core 2 Extreme (X6800)	2.93	1066	2x2 MB	65	291
Core 2 Duo E6700	2.66	1066	2x2 MB	65	291
Core 2 Duo E6600	2.4	1066	2x2 MB	65	291
Core 2 Duo E6400	2.13	1066	2x1 MB	65	167
Core 2 Duo E6300	1.86	1066	2x1 MB	65	167
Pentium D XE965 (Presler Extreme edition)	3.73	1066	2x2 MB	65	376
Pentium D 930 (Presler)	3	800	2x2 MB	65	376
AMD					
AMD 64 FX 62 (AM2)	2.8	2000	2x1 MB	90	227
AMD 64 5000+ (AM2)	2.6	2000	2x512 KB	90	154